KARTHIK CHANDRAN LAKSHMANASAMY

Alexandria, VA | **U.S. Permanent Resident | No Sponsorship Required**
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PROFESSIONAL SUMMARY

Principal Software Engineer in Test with 9+ years of experience leading enterprise quality engineering across SaaS, cloud, distributed, data-intensive, and AI/LLM platforms. Expertise in Python/Pytest automation, UI/API testing, CI/CD, performance and reliability engineering, cloud-native testing, and risk-based test strategy. Proven track record modernizing legacy frameworks, validating RAG/LLM and semantic layer systems, building scalable automation across multi-cloud environments, and mentoring engineers on production-grade quality practices.

CORE EXPERTISE

AI/LLM & RAG Testing | Agentic AI & MCP Testing | Python / Pytest | Selenium / Playwright / WebdriverIO | API & Integration Testing | Performance & Concurrency Testing | CI/CD & Quality Guardrails | Docker / Kubernetes | AWS / Azure / GCP | Distributed Systems | Framework Architecture & Modernization

PROFESSIONAL EXPERIENCE

Principal Software Engineer in Test **May 2023 - Present**
MicroStrategy *Tysons, VA*

- Modernized a 20-year-old legacy automation framework for ETL, SQL, and data validation by migrating core regression workflows to Python/Pytest and containerizing execution with Docker/Kubernetes, improving maintainability and CI/CD integration.
- Developed end-to-end automation frameworks using WebdriverIO and Python for Strategy Mosaic, improving regression coverage across complex enterprise analytics and key data workflows.
- Built automation to validate Mosaic's Universal Semantic Layer, including semantic models, metrics, relationships, ETL transformations, and SQL accuracy across enterprise data sources and BI platforms.
- Automated AI-driven data modeling workflows, including linked models, base/derived metrics, automatic metric generation, and AI-powered analytics.
- Developed automated testing for Model Context Protocol (MCP) integration with Claude, validating connector setup, tool discovery, semantic-model access, query execution, response accuracy, permissions, and execution modes.
- Built an LLM-as-a-Judge framework to evaluate AI-generated responses against Mosaic's governed semantic layer, measuring accuracy, grounding, consistency, hallucination risk, and semantic correctness.
- Implemented predictive and RAG testing using Python and TensorFlow to validate conversational AI and forecast-related responses.
- Built Python automation for cloud deployments across AWS, Azure, GCP, OpenShift, and StackIT, covering provisioning, microservice health, upgrades, autoscaling, concurrency, recovery, and stability.
- Automated pre/post-deployment validation using AWS CodeBuild and designed Jenkins CI/CD workflows for functional, integration, and regression testing.
- Built Python concurrency and performance-testing frameworks for event-driven data models to identify throughput, latency, and scalability bottlenecks.
- Mentored junior engineers on automation architecture, cloud testing, and tooling selection, helping achieve 90% automated test coverage.

Senior Software Engineer in Test **Jul 2019 - May 2023**
MicroStrategy *Poland*

- Designed an automated framework using Protractor, JavaScript, and Cucumber, reducing manual testing effort for the SAP S/4 Connector, cloud-based applications, and services.
- Enhanced Kafka-Flink data stream performance by 38% by optimizing and validating batch size and processing intervals under data-intensive workloads.
- Played a key role in certification and release validation for ADLS, S3, and GCS cloud connectors in MicroStrategy 2021.
- Led FedRAMP certification testing for the Cloudera Kubernetes platform, ensuring compliance with federal security and platform requirements.
- Built a data pipeline to analyze server logs and surface performance metrics used to optimize backend services.

Software Engineer in Test **Mar 2018 - Jul 2019**
MicroStrategy *Tysons, VA*

- Designed a Pytest and Selenium framework to detect Google BigQuery data import issues, improving reliability of data connectivity workflows.
- Streamlined CI/CD workflows by partnering with DevOps to integrate automated tests into pipelines, improving deployment reliability and continuous release validation.
- Managed planning, integration, and continuous testing for the Google BigQuery connector in MicroStrategy.
- Led Teradata-to-Google BigQuery migration validation with cross-functional teams to improve migration quality and data accessibility.

Associate Software Engineer in Test **Mar 2017 - Mar 2018**
MicroStrategy *Tysons, VA*

- Designed a performance testing framework to benchmark big data connectors across large-scale workloads.
- Improved large-scale data import response time by 44% using the Google Storage API and performance-focused validation.
- Led continuous improvement initiatives to increase test efficiency, reduce flakiness, and improve defect detection.
- Collaborated with cross-functional teams to define, review, and prioritize requirements and features.
- Implemented and validated Kerberos authentication for Cloudera Data Platform using Active Directory, strengthening enterprise security.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, Java, Bash, Groovy, C++
Test Automation: Pytest, Selenium, Playwright, WebdriverIO, Cypress, Protractor, Cucumber, JUnit, TestNG
AI / LLM Testing: Generative AI, LLM evaluation, LLM-as-a-Judge, RAG validation, prompt/response grounding, TensorFlow, Azure Embeddings, LangChain, OpenAI, AWS Bedrock, Gemini, Claude, GPT-OSS, MCP
API & Integration: Postman, Swagger, REST/API validation, microservices, data connectors
Performance & Reliability: JMeter, Gatling, concurrency testing, load/performance benchmarking, scalability, regression testing
CI/CD & DevOps: Jenkins, AWS CodeBuild, Git, Docker, Kubernetes, Terraform, Ansible
Cloud & Platforms: AWS, Azure, GCP, OpenShift, StackIT
Data & Distributed Systems: Kafka, Flink, BigQuery, Databricks, Snowflake, Oracle, MySQL, MongoDB, PostgreSQL, Redshift, Redis
BI / Enterprise: MicroStrategy, Tableau, Qlik, Salesforce, ServiceNow, Power BI, ThoughtSpot, Amazon QuickSight

EDUCATION

Master of Science in Computer Science | Texas State University | Aug 2013 - May 2016
Bachelor of Engineering in Computer Science | Anna University | Sep 2009 - May 2013

PUBLICATIONS & PACKAGES

Pawzok - Open-source Python package for testing distributed systems and validating end-to-end data workflows. PyPI | GitHub
Numvo - Open-source Python package for phone-number spam and risk reputation checks. PyPI | GitHub